

Geometry-aware quantum cloud challenge

Hackathon overview

Qiskit Fall Fest 2026

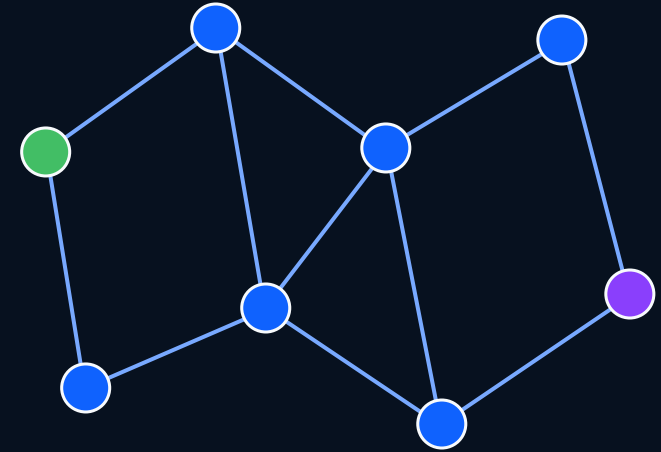
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Connectivity, routing and noise

Quantum circuits often require interactions between qubits that are not directly connected.

Routing introduces extra operations to realize those interactions. Under noise, these operations can reduce the quality of the result.



Two designated ports

quanTA research and the cloud context

Geometry and error protection

quanTA's research connects circuit diagrams and quantum error correction with questions about the geometry of physical implementations.

Hyperbolic superconducting architectures

Finite hyperbolic lattices motivate alternative patterns of connections and cycles. This challenge tests their consequences for routing and error-detection circuits.

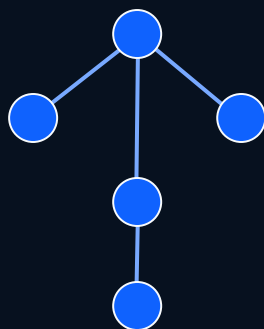
A decade of cloud access

IBM opened its five-qubit cloud processor in 2016. Qiskit now provides a common framework for comparing historical, contemporary and hypothetical processor models.

Processor models across stages

Processor A - Online

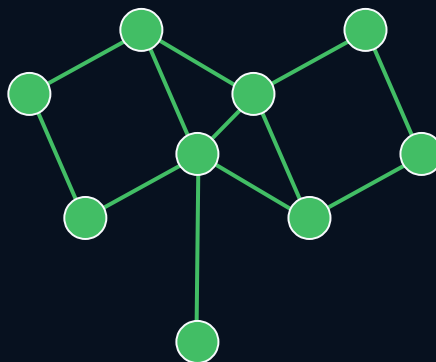
5 qubits



Fixed baseline graph
Synthetic noise model

Processor B - Online

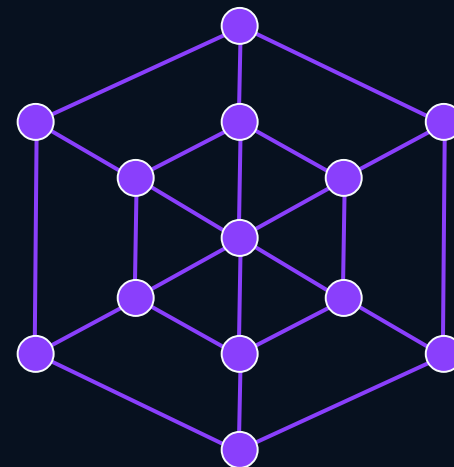
7 qubits



Heavy-hex-inspired graph
Contemporary compilation
baseline

Processor C - Shortlisted

8-10 qubits



Experimental coupling graph
Released after shortlisting
Circuit-level model

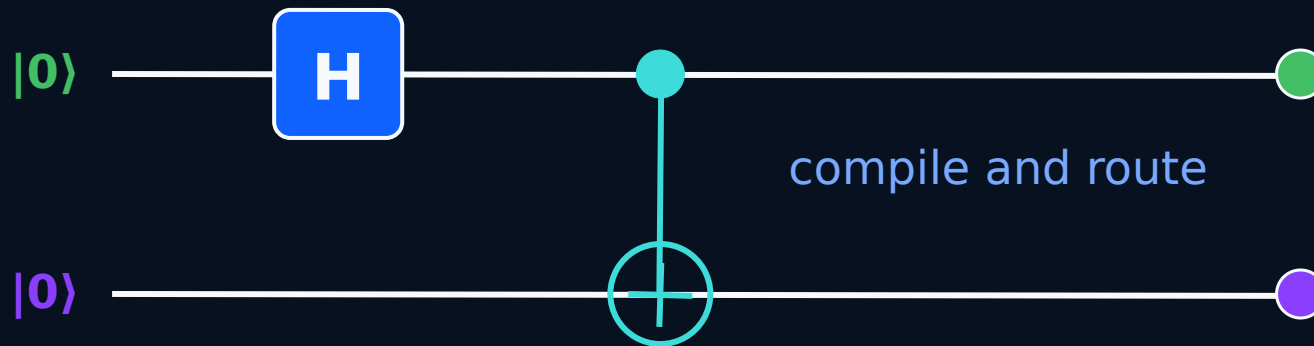
Challenge objective

Online Phase: benchmark your selected solution on Processor A and Processor B.

Build both processors from the provided definitions, run the same controlled experiment, and compare architecture-aware metrics.

After shortlisting, Processor C is introduced and teams design Processor D. The Bell-correlation challenge is completed offline using C and D.

The Bell-correlation task



Verification correlations

XX **YY** **ZZ**

Together, these estimate fidelity with the target Bell state.

The Bell-correlation experiment is reserved for the offline final. Shortlisted teams compare C and D using fidelity, circuit cost, routing overhead and protection results.

Core comparison

Stage / processor	Ideal / reference	Noisy baseline	Final analysis
Processor A - Online	Case A1	Case A2	A/B architecture metrics
Processor B - Online	Case B1	Case B2	A/B architecture metrics
Processor C + D - Offline	Case C/D-1	Case C/D-2	Bell + protection analysis

Online: compare A/B using solution quality, circuit cost, routing and noise-aware architecture metrics. Offline: analyze C/D with Bell correlation and protection overhead.

Protection strategies

Parity or stabilizer checks

Additional measurements can reveal particular errors without directly measuring the encoded information.

Syndrome postselection

Keep outcomes that pass a specified check. Any improvement in conditional fidelity must be reported together with the fraction of accepted shots.

Geometry-aware encodings

More advanced teams can adapt an encoding or check network to the available connections, then measure its overhead.

Challenge rules

Stage the work as follows: online teams use only Processor A and Processor B. Shortlisted teams receive Processor C and design Processor D for the offline challenge.

Use the same selected problem, controlled settings and reporting format on A and B. Submit reproducible results and architecture-aware metrics.

Offline teams run the C/D experiment, perform Bell correlation, and report fidelity, circuit cost, routing overhead and protection results.

Choose the implementation, routing method and analysis strategy for the A/B benchmark.

After shortlisting, choose the Processor D topology and justify it using observations from A, B and the newly introduced C.

Optional studies and QPU results may be included, but must remain separate from the common benchmark.

Required measurements

Depth | two-qubit gates | SWAP overhead | physical qubits | PS-specific output metric

Online A/B benchmark	
Measure	Report
Solution quality	Problem-specific output / reference comparison
Circuit cost	Depth and number of two-qubit gates
Routing	SWAP overhead and connectivity effects
Qubit use	Physical qubits, including ancillas
Noise / reproducibility	Noise impact, assumptions and repeatability

Offline C/D final: add Bell fidelity with uncertainty, XX, YY, ZZ, protection overhead and postselection yield.

Approaches for different experience levels

Experience	Suggested approach	Main comparison
Explorer	Implement the selected PS and compare Processor A vs B.	A/B architecture metrics
Builder	Improve placement, routing or noise-aware execution on A/B.	Fidelity / solution quality versus circuit cost
Research	After shortlisting, study C and design D; analyze Bell correlation offline.	C/D architecture, Bell fidelity and protection overhead

Interpreting the comparison

Noise and compilation

Relate changes in fidelity to the operations introduced by compilation and to the assumed noise model.

Protection overhead

Assess whether the protection method compensates for its extra gates, ancillas and discarded shots.

Architectural conclusions

Distinguish connectivity effects from differences in gate sets or noise. A result from one finite model does not establish a universal advantage.

Required submission materials

01	Phase-1 reproducible notebook	Qiskit code, assumptions, Processor A/B definitions and recorded results
02	A/B graphs and circuits	The two coupling graphs, transpiled circuits and architecture metrics
03	Online quantitative comparison	Problem-specific quality, circuit cost, routing, qubit use and noise-aware results
04	Scientific interpretation	An explanation of the A/B results, limitations and architecture trade-offs
05	Offline final package	C/D design, Bell-correlation results, protection analysis and final technical explanation

Evaluation criteria

Scientific correctness and reproducibility

Correct implementation, reproducible methods and trustworthy results.

Quality of architecture comparison

Evidence-based comparison of the required processor stages.

Protection / Bell-correlation analysis

For the offline final: Bell correlation, protection strategy and overhead.

Effective use of Qiskit

Appropriate use of Qiskit, transpilation and experiment tooling.

Technical interpretation

Clear explanation of results, limitations and engineering implications.

Clarity of presentation

Clear final communication, documentation and technical defense.